Homework 7

Solutions must be **typed** and **submitted electronically**.

1. Let F be a secure PRP with block length $n + \lambda$. Consider the following encryption scheme:

$$\mathcal{K} = \{0,1\}^\lambda \quad \mathcal{M} = \{0,1\}^n \quad \mathcal{C} = \{0,1\}^{n+\lambda}$$

$$\begin{array}{l} \underline{\textsf{KeyGen}:} \\ \quad k \leftarrow \{0,1\}^\lambda \\ \quad \text{return } k \end{array} \qquad \begin{array}{l} \underline{\textsf{Enc}(k,m):} \\ \quad r \leftarrow \{0,1\}^\lambda \\ \quad c := F(k, m \| r) \\ \quad \text{return } c \end{array}$$

   (a) Write the corresponding Dec algorithm.

   (b) Prove that the scheme has CPA$ security.

2. Let F be a secure PRP with blocklength λ. Show that the following construction does not have CPA/CPA$ security:

$$\begin{array}{l} \underline{\textsf{Enc}(k,m):} \\ \quad s_1 \leftarrow \{0,1\}^\lambda \\ \quad s_2 := s_1 \oplus m \\ \quad x := F(k, s_1) \\ \quad y := F(k, s_2) \\ \quad \text{return } (x, y) \end{array}$$